

Patient Readmission Risk & Healthcare Analytics

1. Business Problem

Hospital readmissions within 30 days of discharge are an important healthcare performance metric. Frequent readmissions can indicate that certain patients may require additional follow-up, care coordination, or post-discharge support.

Healthcare organizations need a way to analyze historical patient and hospital encounter data to understand which patient segments have higher observed 30-day readmission rates and which factors are associated with those outcomes.

This project analyzes patient encounter data to identify patterns in 30-day readmissions across demographics, diagnoses, previous hospitalizations, admission characteristics, length of stay, insurance, and discharge information.

2. Business Objective

The objective of this project is to build an interactive healthcare analytics solution that helps stakeholders:

- Measure the overall 30-day patient readmission rate.
 - Identify patient segments with higher observed readmission rates.
 - Analyze the relationship between previous admissions and readmission.
 - Examine readmission patterns across age groups and diagnoses.
 - Understand how admission type, length of stay, and discharge type relate to readmission.
 - Compare readmission rates across insurance categories.
 - Identify patients or patient segments requiring closer analytical attention based on defined risk factors.
 - Provide an interactive dashboard for healthcare management and operational analysis.
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3. Key Business Question

Which patient segments have the highest observed 30-day readmission rates, and which patient and hospital encounter factors are associated with those readmissions?

4. Supporting Business Questions

Patient Demographics

- Which age groups have the highest 30-day readmission rate?
- Does readmission vary across gender groups?
- Which demographic segments contribute the largest number of readmissions?

Clinical Factors

- Which diagnoses have the highest readmission rates?
- Do patients with multiple chronic conditions show different readmission patterns?
- Is a higher number of medications associated with increased readmission?

Hospitalization History

- How does the number of previous admissions relate to 30-day readmission?
- Do frequently hospitalized patients show higher observed readmission rates?

Admission & Discharge

- How does admission type affect readmission patterns?
- Do emergency admissions have different readmission rates compared with elective admissions?
- Which discharge types are associated with higher observed readmission rates?

Length of Stay

- How does length of hospital stay relate to 30-day readmission?
- Which length-of-stay groups have the highest readmission rate?

Follow-up Care

- Is there a difference in readmission rates between patients with and without scheduled follow-up care?

5. Key Performance Indicators

The dashboard will track the following KPIs:

KPI	Description
Total Patients	Number of unique patients analyzed
Total Encounters	Total hospital encounters in the dataset
30-Day Readmissions	Number of encounters followed by readmission within 30 days
Readmission Rate	Percentage of analyzed encounters resulting in 30-day readmission
Average Length of Stay	Average number of days patients stayed in the hospital
Previous Admissions	Average or distribution of prior hospital admissions

KPI	Description
High-Risk Patients	Number of patients classified as high risk using the project-defined analytical scoring framework

6. Risk Segmentation

The project will create an analytical risk segmentation framework using selected historical patient and encounter characteristics.

Potential risk factors include:

- Previous hospital admissions
- Patient age
- Number of chronic conditions
- Length of stay
- Admission type
- Previous readmission history
- Follow-up availability
- Other relevant variables available in the dataset

Patients will be grouped into:

Risk Category Purpose

Low Risk Lower combination of selected historical risk factors

Medium Risk Moderate combination of selected risk factors

High Risk Higher combination of selected risk factors

Important: This risk segmentation is an analytical framework created for this project. It is not intended to represent a clinically validated prediction model or medical diagnosis.

7. Analytical Approach

Step 1 — Data Preparation

- Inspect patient and encounter data.
- Identify missing and inconsistent values.
- Remove duplicate records where appropriate.
- Standardize categorical fields.
- Validate date fields.

- Create derived analytical variables such as age groups and length-of-stay groups.

Step 2 — SQL Analysis

SQL will be used to:

- Calculate overall readmission metrics.
- Analyze readmissions by demographic segments.
- Analyze diagnosis-level readmission patterns.
- Compare admission and discharge types.
- Analyze previous admission history.
- Examine length-of-stay patterns.
- Calculate segment-level readmission rates.
- Prepare datasets for Power BI.

Step 3 — Power BI Analysis

Power BI will be used to:

- Build the data model.
- Create DAX measures.
- Develop KPI cards.
- Create interactive visualizations.
- Build patient risk segmentation views.
- Add filters and drill-down capabilities.
- Present findings through an executive dashboard.

8. Expected Dashboard Pages

Page 1 — Executive Overview

Provides a high-level view of:

- Total patients
- Total encounters
- 30-day readmissions
- Readmission rate
- Average length of stay
- Readmission trends
- Readmission by age and diagnosis

Page 2 — Patient Risk Analysis

Focuses on:

- Low / Medium / High risk distribution
- Risk by age group
- Risk by previous admissions
- Risk by diagnosis
- Risk by length of stay
- High-risk patient segments

Page 3 — Clinical & Operational Analysis

Focuses on:

- Admission type
- Discharge type
- Insurance
- Follow-up scheduling
- Length of stay
- Diagnosis-level patterns

9. Expected Business Value

The analysis is intended to help healthcare stakeholders better understand historical readmission patterns and identify patient segments that may warrant closer review.

Potential applications include:

- Supporting post-discharge care planning.
- Identifying patient groups for additional follow-up analysis.
- Understanding operational patterns associated with readmissions.
- Monitoring readmission KPIs.
- Supporting data-driven healthcare management decisions.

The dashboard provides analytical insights from historical data and should not be used as a standalone clinical decision-making or patient-treatment system.

10. Technology Stack

Data Analysis

- SQL

- PostgreSQL / MySQL

Business Intelligence

- Microsoft Power BI
- DAX
- Power Query

Optional

- Python
- Pandas
- Matplotlib / Seaborn

Documentation

- GitHub
 - Markdown
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11. Project Deliverables

The completed project will include:

1. Business Problem Report
 2. Dataset Documentation
 3. Data Cleaning Process
 4. SQL Analysis Scripts
 5. Data Dictionary
 6. Power BI Data Model
 7. DAX Measures
 8. Interactive Power BI Dashboard
 9. Key Findings
 10. Business Recommendations
 11. Project README
 12. Screenshots of the final dashboard
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12. Success Criteria

The project will be considered successful if the final solution can clearly answer:

How frequently are patients readmitted within 30 days?

Which patient and encounter segments show higher observed readmission rates?

Which factors are associated with higher readmission rates in the analyzed data?

Which patient segments fall into the project's defined higher-risk categories?

Can healthcare stakeholders interactively explore these patterns through a Power BI dashboard?

13. Project Scope

In Scope

- Historical patient encounter analysis
- 30-day readmission analysis
- Patient segmentation
- Demographic analysis
- Diagnosis analysis
- Admission and discharge analysis
- Length-of-stay analysis
- Insurance analysis
- Follow-up analysis
- Analytical risk segmentation
- SQL analysis
- Power BI dashboard

Out of Scope

- Clinical diagnosis
 - Treatment recommendations
 - Individual medical advice
 - Clinically validated readmission prediction
 - Automated patient treatment decisions
 - Real-time hospital monitoring
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14. Final Business Outcome

The final solution will transform raw patient encounter data into an interactive healthcare analytics dashboard that enables stakeholders to explore **30-day readmission patterns, identify higher-risk patient segments based on the project's analytical framework, and understand the operational and patient-level factors associated with readmissions.**

